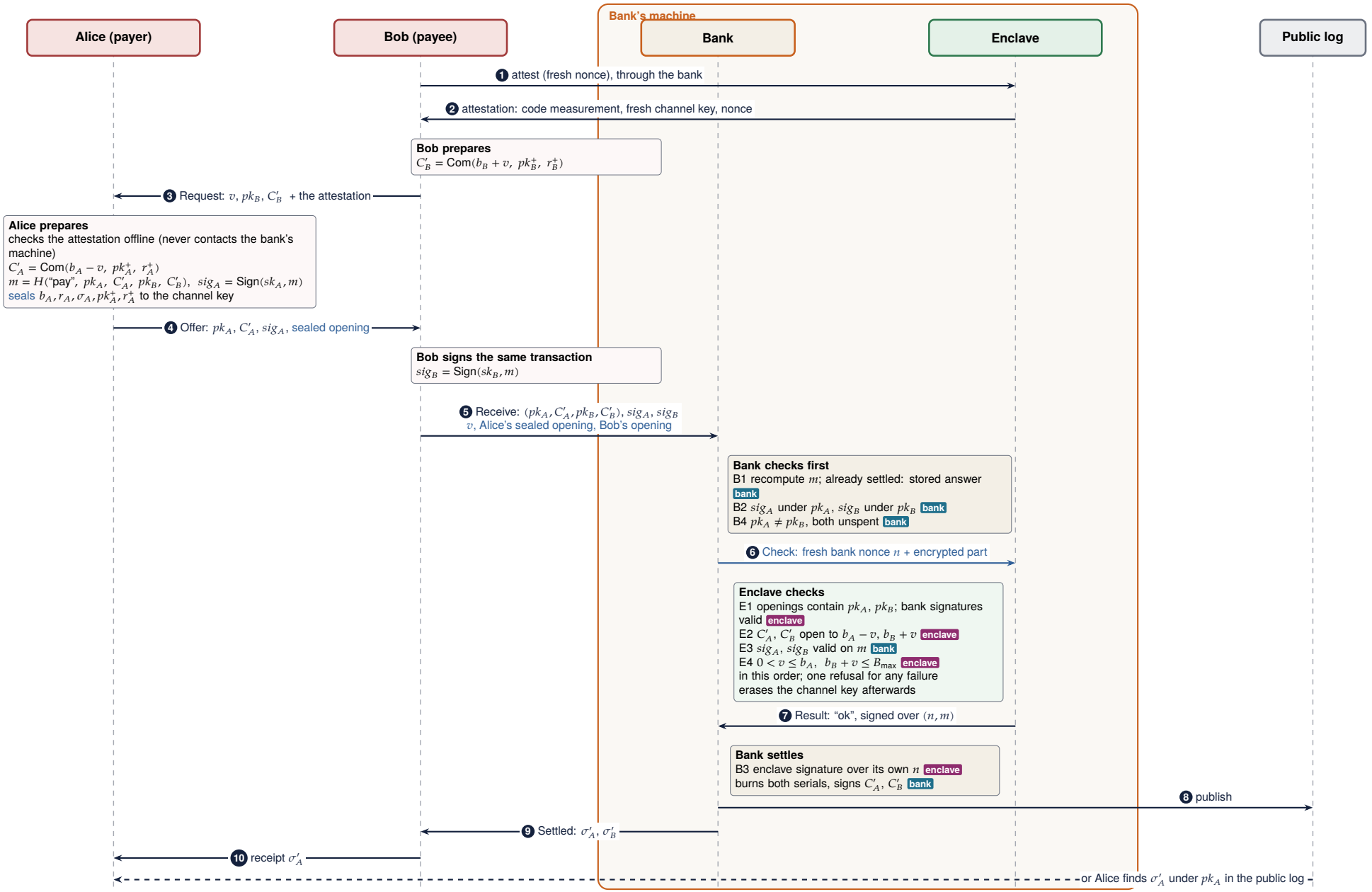
Round 1: a payment

Alice pays v to Bob; only Bob contacts the system

blue encrypted to the attested enclave bank check the bank does itself enclave rests on the enclave, accepted over the bank's own channel
$^+$ next value, for example pk_A^+ is Alice's next serial key $\mathrm{Com}(b, pk, r)$ hides a state, σ is the bank's signature on it



Why both owners sign $m = H(\text{"pay"}, pk_A, C'_A, pk_B, C'_B)$**.** Each serial is a one-time public key, so only the owner of a state can spend it, and the bank checks this itself (B2): signing keys never reach the enclave. The two signed commitments fix both new balances, so the amount and the payee cannot be changed. v is left out of m so the bank can recompute m without learning the amount. **What stays on the enclave:** that the spent states exist and that balances are conserved (E1, E2, E4), which the bank cannot see. The bank accepts that answer only from its own enclave, signed over a nonce it just chose, so nobody outside the bank's machine can submit a result.